

# A Categorization Model of Requirements Based on Max-Neef's Model of Human Needs

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## ABSTRACT

Requirements categorization is an inherent part of the requirements engineering activity. Conventional approaches use a designer perspective (requirements organized according to design needs or attributes), a contractual perspective (requirements organized according to procurement or acquisition needs), or a combination of both. Such models present several inconveniences that result in limitation of system affordability: facilitate the generation of overlapping requirements, of design-dependent requirements, and of a mix of requirements applicable to different levels of the architecture decomposition or to different products. The present research proposes a Need-based Categorization (NbC) model that is system-centric: Requirements are organized around the system. Inspired by Max-Neef's model of human needs, the proposed model supports requirement elicitation by defining only what the system does, how well, where, and what it uses to accomplish it. The model facilitates the identification of constraints that limit the solution tradespace without supporting the satisfaction of new needs, of overlapping requirements, and of requirements that are not applicable to the system. Finally, the proposed model defines requirements in subsets that are associated with value to stakeholders, thus reflecting the actual dependency nature of requirements at a given level of an architecture decomposition, which promotes holistic decisions instead of local optimizations. © 2013 Wiley Periodicals, Inc. *Syst Eng* 17: 348–360, 2014

Key words: requirements categorization; requirements prioritization; system analysis and design; stakeholder needs

## 1. INTRODUCTION

Requirements engineering is considered a critical step in developing successful systems [Verner et al., 2007], and, in fact, it is seen by some authors as “the cornerstone of the

systems engineering process” [Buede, 2009: 151]. Because “it is not possible to have an acceptable system even with the best solution space if this is based on an incorrect problem space formulation” [Brace and Thramboulidis, 2010], the majority of systems engineering standard processes use requirements engineering as one of the first steps in the development of a system [INCOSE, 2011; NASA, 2007].

Several researchers have explored the effects requirements have on system failure. Weinberg [1997] concludes that up to 60% of errors have their origin in the requirements engineering process. Lyytinen and Hirschheim [1987] concluded that 25% to 90% of system failures were caused by one or more problems derived from requirements: The system does not correspond to what was required, the system has cost and

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schedule overruns, the system failed to satisfy users, or the system was unable to satisfy stakeholder needs. Yeo [2002] and Dada [2006] corroborated such results in more recent investigations. Correlation between requirement practices and productivity has also been confirmed in an extensive survey spanning 56 projects within 29 organizations over 2 years [El Emam and Birk, 2000]. In particular, requirements can play a crucial role in reducing cost of change since a factor between 10 and 200 has been measured for cost of change during implementation phase with respect to the requirement elicitation phase [Boehm and Papaccio, 1988; McConnell, 2001]. It is thus not surprising that industry and academia seek and propose measures to improve existing capabilities in requirements engineering [Hooks, 2001; INCOSE, 2012; Niu et al., 2011; Kaindl et al., 2002]; yet its application does not seem to convey the required quality [Hooks, 1990, 2001, 2010; Kaindl et al., 2002; Fanmuy and Foughali, 2012], and complex systems continue to fail [Bahill and Henderson, 2005; Schwenn et al., 2011].

Our research in this paper is grounded on the idea that existing categorization models of requirements are only used to define the problem boundaries or the system under development, consequently passively or reactively influencing system affordability. Furthermore, the approach or perspective under which existing categorization models of requirements are usually developed and used, i.e., centered on the designer or on contractual obligations, ultimately results in limited system affordability due to an overconstrained solution tradespace. As a result we propose that requirements are actively used to drive system affordability instead. To achieve such a goal we suggest a categorization model that facilitates the maximization of the solution tradespace for a given set of stakeholder needs instead of its mere definition. This model would (1) determine the right requirements right and (2) determine only the minimum necessary requirements for a given set of stakeholder needs.

Because “[e]very requirement costs” [Arnold and Martin, 2005], the present research proposes a cognitive categorization model to organize requirements inspired by Max-Neef’s work and research on human needs, in particular on his proposition of existential categories [Max-Neef, 1989]. This model supports requirement elicitation by reducing the amount of requirements to the minimum necessary to satisfy a given set of stakeholder needs. In addition, it realistically captures the perceived value of subsets of requirements to stakeholders. Using a system perspective, we propose that this model is able to detect requirements that influence the design as constraints and those that are related to how the system shall be procured as non-system requirements artifacts.

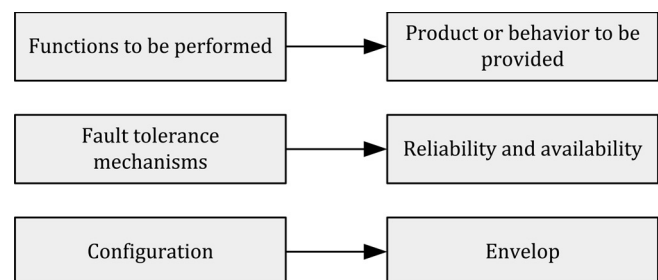
This paper is organized as follows. Section 2 provides a review of existing literature on the topics of categorization and prioritization of requirements, capture of stakeholder value, and psychological models of human needs. Section 3 presents the proposed Need-based Categorization (NbC) model for requirements and its application is described in Section 4. The paper finishes giving a short summary of the conclusions and a proposal for future work in Section 5.

## 2. LITERATURE REVIEW

### 2.1. Categorization of Requirements

Categorization of requirements is a common practice in systems engineering to support requirement elicitation that aims at completeness [Buede, 2009], to easily manage their control [Hull et al., 2005], and to establish a kind of priority to guide designers and architects in finding and developing solutions to defined problems [Jain et al., 2008]. Jain et al. [2008] use requirement type to evaluate the impact on system integration complexity.

NASA [2007] establishes several categories of requirements grouped in five areas that are in principle tailored to a standard space project: Technical Requirements (Functional Requirements, Performance requirements, and Interface Requirements), Operational Requirements (Mission, Configuration, and Command and Telemetry), Reliability Requirements (Environment, Fault tolerance, Verification, and Process and Workmanship), Safety Requirements, and Specialty Requirements (Maintainability, Producibility, etc.). The European Space Agency follows a similar approach: Functional Requirements, Mission Requirements, Interface Requirements, Environmental Requirements, Operational Requirements, Human Factor Requirements, (Integrated) Logistics Support Requirements, Physical Requirements, Product Assurance Requirements, Configuration Requirements, Design Requirements, and Verification Requirements [ECSS, 2009]. Such type of classification of requirements facilitates the specification of how the system has to be designed, which implies influencing the solution, whereas it should rather specify what the system is intended to do. For example, as shown in Figure 1, *Fault Tolerance Requirements* promote the specification of fault tolerance mechanisms instead of reliability and availability requirements. Similarly, *Configuration Requirements* overconstraint envelope requirements and specifying functions limits the amount of functional or physical architectures that could provide the same behavior or product. In addition, these definitions of categories do not fulfill the partition criterion and therefore facilitate the emergence of overlapping requirements; e.g., there is no clear delimitation between functional and interface requirements, mission and environment requirements, or producibility and process requirements.



**Figure 1.** Conventional design specifications vs. system requirements.

In the medical industry Ward et al. [2003] propose a matrix classification based on 5 domains (process, performance, safety, cost, and documentation) and 7 stages of the system life-cycle (design, manufacturing, distribution, installation, operation, maintenance, and disposal), ending up with 35 categories. Again in this case, the authors generate requirements not to the system, but to the project or how the system is acquired, such as documentation or process requirements.

The International Council of Systems Engineering (INCOSE) [2011] proposes an independent classification of requirements that targets any complex system and that includes functional requirements, performance requirements, non-functional requirements, and architectural constraints. Hull et al. [2005] make a similar contribution in the field of software systems and define functional, performance, quality factor, environment, interface, and constraint as requirements types. Both categories provide excessive freedom in some of the categories such as nonfunctional requirements or quality factor and facilitate or at least invite definition of constraints; yet we find it valuable that these are identified in a stand-alone group so that their impact on the system can be easily evaluated. In an effort to conceptually identify generic types of requirements, Bapat et al. [2007] propose that requirements exist on design attributes, on the existence of objects and characteristics, on the relationships, and on functions. Their proposition confirms a designer perspective when eliciting requirements: How the system has to be designed. In an intent to provide requirements with engineering formulation, Gilb [1997] proposes a categorization of requirements that define the elements a designer would work with to develop a system, namely: function requirements, which indicate what the system must do; performance requirements, which define how well the functions of the system must perform; resource requirements, which define the resources that are available to create and maintain the functions and performance of the system (explicitly referring only to money, people, and time);

design constraints, which define restrictions on the solution; and condition constraints, which define restrictions on the use of the system.

Together with his mathematical proposition to systems engineering, Wymore [1993] makes the first attempt to set a categorization of requirements that can completely define a system independently of its intended use, thus providing a domain-independent classification: Input/Output requirements, Technology requirements, Performance requirements, Cost requirements, Trade-off requirements, and System Test requirements. Buede [2009] builds on his work and reorganizes them hierarchically while complementing it with additional types of requirements, as shown in Figure 2.

Both propositions isolate constraints under dedicated categories and thus move almost entirely away from a designer perspective. They are constructed under the principle of partition from set theory, and therefore aim at minimizing the amount of necessary requirements by eliminating overlapping requirements while ensuring the system is completely specified. However, their classifications still include artifacts that do not really belong to the requirements of a system and therefore they do not manage to achieve a concise and complete decoupling between boundaries and solution. First of all, both propositions are presented as system design requirements, thus implying a relation between requirements and the actual design, whereas the present research proposes requirements should be linked to the system instead. Moreover, *Trade-off* requirements can be understood as a representation of value to stakeholders and therefore they are attributes of the system requirements, but not requirements on their own merit. In addition, the proposed *Trade-off* requirements overlap with cost and schedule requirements, which jeopardizes the concept of partition. The same is valid for functional and input/output requirements. Finally, a similar reasoning can be applied to the System Test Requirement of System Qualification Requirements category, since they involve contractual

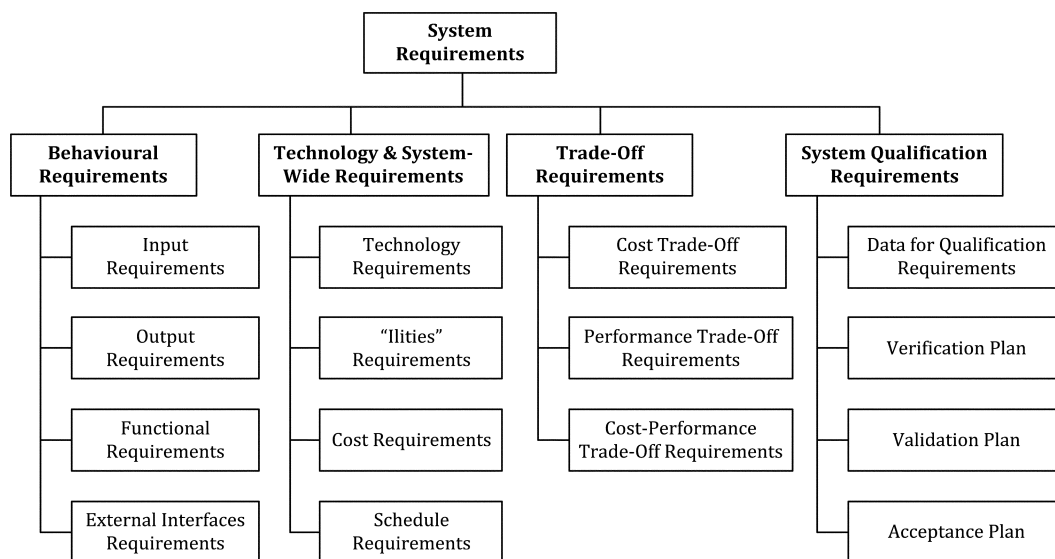


Figure 2. Buede's classification of requirements [Buede, 2009].

agreements and understanding of system requirements: the way a system is verified or validated does not affect or define what the system is.

As it has been elaborated in this section, the way in which requirements are categorized can impact system affordability, yet scarce research has investigated how to develop a classification that provides added value to the elicitation process. As a matter of fact, the majority of existing categorizations of requirements have been traditionally developed following a designer perspective, by thinking about *what needs to be there to completely design a system*, or a contractual perspective, by thinking about *what needs to be delivered as part of the contract*. The designer perspective facilitates a writing style that provides requirements for the design of the system and not for the system itself, thus limiting the solution tradespace for the system. The contractual perspective specifies how to procure the system, and thus it changes the scope of requirements. Basically none of those classifications achieves a proper description of what the system is intended to do, and instead they end up consistently increasing the amount of requirements without satisfying any new stakeholder need. Standardization processes led to a status quo that has been simply assumed by the engineering community.

## 2.2. Prioritization of Requirements

Firesmith [2004] provides an exhaustive list of prioritization techniques: business case analysis/Return On Investment (ROI) estimation, Pairwise comparisons, Prioritization working groups, Scale of 1-to-10 rankings, Voting schemes, Weightings, Value-Based Software Engineering [Boehm, 2003], WIN-WIN [Boehm et al., 2001], and Quality Function Deployment (QFD) [Akao, 1992].

Brackett [1990] proposes to divide requirements into three qualitative levels of importance, namely, *mandatory*, *desirable*, and *essential*. Karlsson [1995] extended such concept to provide granularity on the relative importance by assigning a number from 1 to 5 depending on the importance of each requirement, namely, *does not matter*, *not important*, *rather important*, *very important*, and *mandatory*. Evaluation of importance is assessed by averaging the ranking of different stakeholders. Such methods are known as Numeral Assignment Technique [Mead, 2008]. Leffingwell and Widrig [2003] propose the 100-Point Method in which the different stakeholders are given 100 points to distribute among the requirements. Averaging between the “bets” of the different stakeholder is used to assess the final priority of the different requirements.

In the WIN-WIN method the key objective is to solve disagreement between different stakeholders [Boehm et al., 2001; Boehm and Ross, 1989; Park et al., 1999]. Basically stakeholders assign preferences with an isolated assessment. Once put in common, disagreements are solved in a take-give approach.

Davis [2005] proposes a different approach to handle requirement prioritization. In his *Requirement Triage*, relative priorities are established for the requirements, and resources needed for their achievement are assessed. Then requirements are packed in subsets, and each subset is evaluated against the probability of such product being a success in the market.

Wiegiers [2003] introduces the impacts associated with the fulfillment of a requirement. In the Wiegiers' method, requirement priority is assessed by relating the value of that requirement to a stakeholder and the cost and risk associated with its development.

The Analytic Hierarchy Process (AHP) can be used for prioritizing requirements with multiple objectives [Saaty, 1980]: value to several stakeholders, importance to the company, risk, or any other type of importance as previously identified. With this method pairwise comparisons between different requirements and weightings between different types of importance (objectives) are performed.

Palma et al. [2011] include requirement dependency as a variable when determining requirement importance. However, it only addresses dependencies for identifying implementation order. The question of how to deal with change is left unaddressed. The importance of managing dependency is also addressed by Carlshamre et al. [2001], who determine that 20% of the requirements are responsible for around 75% of dependencies in a software product.

Despite the high amount of available prioritization techniques, their actual use in industry for hardware-intensive systems is scarce [Fanmuy and Foughali, 2012; Fellows and Hooks, 1998]. Ma [2009] evaluates the effectiveness of some of these techniques in real projects and concludes that they are mostly ineffective for systems with over 50–100 requirements mainly because they eventually become unmanageable due to their incremental complexity. In a similar line of reasoning, Karlsson and Ryan [1997] claimed a lack of industry buy-in for requirement prioritization due to the unavailability of simple, effective, and industrially proven techniques. Moreover, Lubars et al. [1993] concluded that companies in fact did not really know how to prioritize requirements effectively. These results are supported by more recent research, where the actual practice of requirement prioritization in industry is investigated and the following conclusions are drawn [Lethola et al., 2004: 501]:

- “Requirement prioritization is an ambiguous concept.
- Prioritization techniques are informal and dependent on individuals.
- Requirements are prioritized in many phases.
- Developers do not know enough about customer preferences.
- The priority of a requirement is based on many factors.”

## 2.3. Capturing Stakeholder Value

Stakeholder value can be seen as a type of importance [Firesmith, 2004]. However, capturing stakeholder value should not be confused with establishing priorities among requirements in relation to their relative importance to stakeholders; it is something more. Capturing stakeholder value is founded on two ideas that are used in combination for systems with more than one requirement:

1. Determine relative priorities between different requirements with respect to their nominal fulfillment level.

2. Establish within a single requirement the value of different levels of fulfillment for a stakeholder (e.g., speed from 100 to 220 km/h).

Determining relative priorities can be done with several of the techniques presented in the previous section. Capturing the value of different levels of fulfillment of a requirement for a stakeholder (not its elicitation) has been less explored. Two theories are primarily proposed and used: utility theory [Swalm, 1966] and prospect theory [Kahneman and Tversky, 1979].

In utility theory, stakeholder value is represented by a normalized absolute relation between the possible levels of fulfillment of a requirement and the perceived value to the stakeholder, which can take several forms, such as cost or benefit, for example. Figure 3 shows an example of a utility function for the amount of sugar in a coffee: different levels of sugar make a coffee have different value to the customer.

In prospect theory, stakeholder value is represented by a relative relation between a gain or a loss on the level of a requirement with respect to a status quo or expected value and the associated gain or loss in value, which can take several forms, such as cost or benefit, for example. Figure 4 provides an example of applying such theory.

However, such approaches do not provide a solution to tradeoffs that combine both perspectives: decision between requirements at a nonnominal fulfillment level. In addition, value to stakeholders is holistic, in the sense that it is represented by an image of the system and not by the fulfillment of a particular requirement. Consequently, applying utility or prospect theories to individual requirements would not reflect stakeholder reality and would also be unmanageable in practice due to time and economic boundaries of complex projects.

## 2.4. Psychological Models of Human Needs

The present research proposes a categorization model of requirement with basis on psychological models of human needs. Such proposal is grounded on the following argumentation:

1. Psychological models have been more advanced and successful in classifying and modeling human needs.
2. In essence, a human can be considered a complex and sophisticated system, consequently society a generic system.

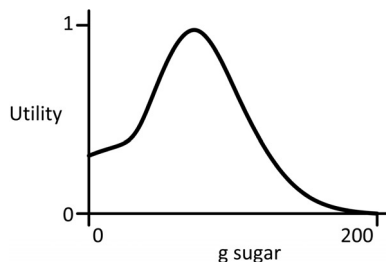


Figure 3. Utility function example for sugar in a coffee.

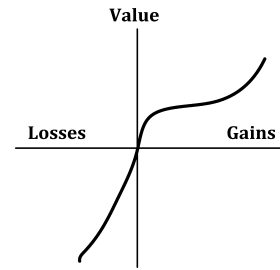


Figure 4. Example of a prospect theory function.

3. Consequently, models representing the classification of human needs could be transported to represent the classification of a system's needs.

Adoption of models of human needs for different uses in several other domains has been significantly successful, especially Maslow's hierarchy of human needs [Maslow, 1943]. Despite their massive use in the domains of team motivation, team management, and the like, they have found considerable success in describing user needs for technology development [Fitzpatrick, 2009; Henry, 2000] and service value [Nakamura et al., 2007; Walsh, 2011].

Maslow [1943] proposed that the closer to a basic need an act or desire is, the more important it is. However, the hierarchy and the grouping are not rigid (see Fig. 5):

- Elements on the pyramid can be swapped depending on culture, religion, etc.
- Full satisfaction of a level is not necessary so that a human seeks and gets satisfaction of higher level needs.

In the same article Maslow also identifies the preconditions that are needed for the basic need satisfaction, for example, freedom to speak, justice, or fairness. In a later publication, he proposed the B-values, which were present in manners of thought that led to peak experiences [Maslow, 1998]; for example:

- *Wholeness*: unity, integration, tendency to oneness, interconnectedness, simplicity, organization, structure, dichotomy-transcendence, order.

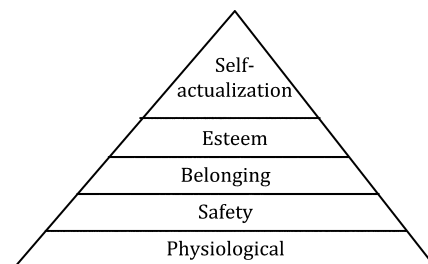


Figure 5. Maslow's pyramid of human needs.

- *Perfection*: necessity, just-rightness, just-so-ness, inevitability, suitability, justice, completeness, ought-ness.
- *Completion*: ending, finality, justice, fulfillment, finis and telos, destiny, fate.

Alderfer [1969] further grouped the five categories of needs identified by Maslow into three: Existence, which combines Physiological and Safety needs; Relatedness, which combines Interpersonal Love and Esteem needs; and Growth, which combines Actualization and Self-Esteem needs. He also proposed that humans redouble efforts in one level when needs in a higher category are not met. Doyal and Gough [1991] made a comparison between human and market needs that supports the problematic of decision analysis with non-monetary benefits:

- Human needs are complex and multidimensional.
- Market needs are simple: expand the earnings managing risk measured in a single dimension, money.

The authors contextualize human needs in society and define them as the “cost of being human” within society. This means that a person shall fulfill their needs in order to function properly within societies. Along these lines, they also mention the need for social preconditions so that needs can be satisfied, because needs exist within a social context.

Max-Neef [1989] augmented Maslow's proposition and incorporated a second dimension to his classification of human needs. Max-Neef [1989] proposed a matrix in which human (axiological) needs encounter four existential categories that define humans as a whole, namely, *Doing*, *Having*, *Interacting*, and *Being*. Human needs satisfaction is then facilitated by the achievement or provision of a combination of satisfiers that are related to each existential category. In Max-Neef's proposition axiological categories are not hierarchical and their relative priorities depend on culture and groups (see Table I).

**Table I. Max-Neef's Model of Human Needs [Max-Neef, 1989]**

| Needs according to axiological categories | Needs according to existential categories                                                                                   |                                                                                                                     |                                                                                                                         |                                                                                                                  |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|                                           | Being                                                                                                                       | Having                                                                                                              | Doing                                                                                                                   | Interacting                                                                                                      |
| Subsistence                               | Physical health, mental health, equilibrium, sense of humour, adaptability                                                  | Food, shelter, work                                                                                                 | Feed, procreate, rest, work                                                                                             | Living environment, social setting                                                                               |
| Protection                                | Care, adaptability, autonomy, equilibrium, solidarity                                                                       | Insurance systems, savings, social security, health systems, rights, family, work                                   | Co-operate, prevent, plan, take care of, cure, help                                                                     | Living space, social environment, dwelling                                                                       |
| Affection                                 | Self-esteem, solidarity, respect, tolerance, generosity, receptiveness, passion, determination, sensuality, sense of humour | Friendships, family, partnerships, relationships with nature                                                        | Make love, caress, express emotions, share, take care of, cultivate, appreciate                                         | Privacy, intimacy, home, spaces of togetherness                                                                  |
| Understanding                             | Critical conscience, receptiveness, curiosity, astonishment, discipline, intuition, rationality                             | Literature, teachers, method, educational policies, communication policies                                          | Investigate, study, experiment, educate, analyse, mediate                                                               | Settings of formative interaction, schools, universities, academies, groups, communities, family                 |
| Participation                             | Adaptability, receptiveness, solidarity, willingness, determination, dedication, respect, passion, sense of humour          | Rights, responsibilities, duties, privileges, work                                                                  | Become affiliated, co-operate, propose, share, dissent, obey, interact, agree on, express opinions                      | Settings of participative interaction, parties, associations, churches, communities, neighborhoods, family       |
| Leisure                                   | Curiosity, receptiveness, imagination, recklessness, sense of humour, tranquility, sensuality                               | Games, spectacles, clubs, parties, peace of mind                                                                    | Day-dream, brood, dream, recall old times, give way to fantasies, remember, relax, have fun, play                       | Privacy, intimacy, spaces of closeness, free time, surroundings, landscapes                                      |
| Creation                                  | Passion, determination, intuition, imagination, boldness, rationality, autonomy, inventiveness, curiosity                   | Abilities, skills, method, work                                                                                     | Work, invent, build, design, compose, interpret                                                                         | Productive and feedback settings, workshops, cultural groups, audiences, spaces for expression, temporal freedom |
| Identity                                  | Sense of belonging, consistency, differentiation, self-esteem, assertiveness                                                | Symbols, language, religions, habits, customs, reference, groups, sexuality, values, norms, historical memory, work | Commit oneself, integrate oneself, confront, decide on, get to know oneself, recognize oneself, actualize oneself, grow | Social rhythms, everyday settings, settings which one belong to, maturation stages                               |
| Freedom                                   | Autonomy, self-esteem, determination, passion, assertiveness, open-mindedness, boldness, rebelliousness, tolerance          | Equal rights                                                                                                        | Dissent, choose, be different from, run risks, develop awareness, commit oneself, disobey                               | Temporal and spatial plasticity                                                                                  |

### 3. A NEED-BASED CATEGORIZATION OF REQUIREMENTS

#### 3.1. Purpose and Objectives

With the purpose of supporting elicitation and prioritization techniques so that (1) the right requirements are rightly elicited, (2) the solution tradespace is maximized while satisfaction of stakeholder needs is maintained, and (3) stakeholder value is realistically reflected and usable, the present research proposes a system-centric categorization of requirements based on the concept of partition in set theory that achieves several objectives:

1. Completely define a system; i.e., all elements necessary to define what the system is intended to achieve shall fit within the proposed categories.
2. Identify requirements that are not applicable to the system to be developed, but to ancillary elements, such as supporting products or contractual agreements.
3. Identify constraints that do not support the satisfaction of stakeholder needs but that limit the solution tradespace, thus facilitating the definition of boundaries for the solution and eliminating any influence on a specific solution.
4. Ease identification of stakeholder value that enables commercial and contractual leverage linked to incrementally achieved value.

In addition, the proposed categorization has been developed aiming at facilitating adoption by practitioners and consequently (1) is compatible with document-centric and model-centric techniques (facilitating transition from one to another), (2) is compatible with and can be supported by existing elicitation and prioritization techniques, and (3) is compatible with and can be supported by existing techniques for capturing stakeholder value.

#### 3.2. From Psychology to Requirements Engineering: The Categorization Model

The present research proposes an adoption of Max-Neef's classification of human needs to a classification of requirements. Requirements in artificial and man-made systems have strong mapping to human needs, and therefore a model inspired by human needs can be of immense use in categorizing requirements in man-made systems. We propose that requirements are organized following Max-Neef's proposition of existential categories and grouped according to their value to stakeholders. Table II exemplifies the proposed categorization.

The use of Max-Neef's existential categories ensures that a system is completely defined by specifying what systems do, how they are (how well they do), what they use, and where they live. Axiological needs ensure stakeholder needs (and consequently requirements) do not have uniform value. Under these terms, human needs are reflected by stakeholder values or need levels, existential categories are reflected by the types of requirements that completely define a system, and satisfiers are reflected by system requirements. With this

Table II. Initial Classification Model Example

| Value level | Functions (Do)             | Performance (Being) | Resources (Have)   | Interaction (Interact) |
|-------------|----------------------------|---------------------|--------------------|------------------------|
| Break-even  | Req. 1<br>Req. 2<br>Req. 3 | Req. 4              | Req. 5<br>Req. 6   | Req. 7                 |
| Goal        | Req. 8                     | Req. 9              |                    |                        |
| Wish        |                            | Req. 10             | Req. 11<br>Req. 12 |                        |

approach, this research proposes the concept of system existential categories, which correspond to the following requirement types:

- **Functional requirements (Do):** Requirements that define what the system must do in essence, or, in other words, what it accepts and what it delivers (i.e., expected transformation). Examples: *The system shall accept coins; The system shall transmit 4 TV channels; The system shall convert sea water into drinkable water.*
- **Performance requirements (Being):** Requirements that define how well the system must operate, which includes performance related to functions the system performs or characteristics of the system on their own, such as -ilities. Examples: *The system shall move at a speed higher than 20 km/h; The system shall have a reliability better than 0.75.*
- **Resource requirements (Have):** Requirements that define what the system can use to transform what it accepts in what it delivers. Examples: *The system shall consume less than 200 W; The system shall have a mass of less than 20 kg.*
- **Interaction requirements (Interact):** Requirements that define where the system must operate, which includes any type of operation during its life-cycle. Examples: *The system shall withstand shock levels higher than 100 g; The system shall operate in vacuum* (to reflect operation); *The system shall operate in clean room class 10,000* (to reflect Assembly, Integration, and Test activities).

In order to ensure adoption and usability, and in order to reflect the reality of requirement interdependencies, requirements are organized in subsets that hierarchically map value to stakeholders. The hierarchical description deviates from Max-Neef's proposition, but serves better contractual purposes by ensuring each level represents a fully operable system as they are options that build upon previous need levels. The amount of levels is unlimited and free of preconceptions, being therefore up to each project to define theirs. For this particular research three levels have been used:

- **Break-even:** the minimum level of service (value) that must be provided so that the system is acceptable.
- **Goal:** desired value provided by the system.
- **Wish:** great-to-have, but considered difficult to achieve.

### 3.3. Solution Constraints: Where Do They Fit?

The present research proposes to allocate solution constraints to the *Resource* category. This is because in essence resources and constraints have a similar effect into the system (both are elements the system uses for transforming its inputs into outputs), yet their behavior is different because resources are available to the system whereas constraints must be implemented. In addition, because constraints do not directly support the satisfaction of needs, they could also be used for trading purposes.

### 3.4. Cost and Schedule: Where Do They Fit?

Cost and schedule limit the solution tradespace and as such traditional categorization of requirements include them as requirements or constraints [Buede, 2009; INCOSE, 2011; NASA, 2007; ECSS, 2009]. Furthermore, they form contractual bonds between the customer and the manufacturing organization. In addition, cost requirements can also be defined for the operational phase of the system, i.e., how much money is required to operate the system at the specified performance levels.

Under these terms, operational cost requirements inherently belong to the resource category, as it is something a system uses to fulfill its functions. Categorization of schedule and development costs is not trivial.

In a first perspective, development cost and schedule can be perceived as resources because they are consumed during system development. However, these resources are not consumed during development by the system, but by the project developing the system, and therefore they would actually reflect project constraints and not system ones. On the other hand, it could also be argued that time and cost are indeed consumed by the system during its creation, which would bring them back as resource requirements to the system.

Another perspective to look at this problem is considering financials. A system eventually becomes an asset with a value that is linked to its development costs once it is transferred to its customer (stakeholder making the expenditure). This viewpoint would imply that cost would not be a resource requirement, but a performance requirement in the form of *The system shall have a value lower than 100 Meuro*. Physics brings a similar idea. The cost and time to develop a system could be understood as energy that is being stored within the system, and that is later consumed during the lifetime of the system (depreciation in economic terms). Again in this case the requirement would fall into the performance category and not into the resource one.

Consequently, the present research proposes to allocate development cost and schedule requirements in one of the following two categories, depending on the vision and needs of each project:

- Resources consumed by the system during its development.
- Performance requirements that represent the *value* of the system.

**Table III. Sample Requirements Categorized According to ECSS [2009]**

| Categ.                     | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mission reqs.              | <b>Req. 1.</b> The instrument shall take images of the Earth surface and transmit them to ground.<br><b>Req. 2.</b> The instrument shall be compatible with a Low Earth Orbit (LEO).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Operat. reqs.              | <b>Req. 3.</b> The instrument shall be powered on by the platform.<br><b>Req. 4.</b> The instrument shall be depressurized during its ascent phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Funct. reqs.               | <b>Req. 5.</b> The instrument shall image the Earth surface in UV spectral range.<br><b>Req. 6.</b> The instrument shall image the Earth surface in VNIR spectral range.<br><b>Req. 7.</b> The instrument shall move a mirror to select the redundant channel.<br><b>Req. 8.</b> The instrument shall power itself.<br>Note: when receiving power according to Req. 12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Interface reqs.            | <b>Req. 9.</b> The instrument shall transmit image data according to IF-Def-1.<br><b>Req. 10.</b> The instrument shall be compatible with the platform payload carrying capability.<br><b>Req. 11.</b> The instrument shall have a thermal coupling at its mechanical interface lower than 2 units.<br><b>Req. 12.</b> The instrument shall accept power according to IF-Def-2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Perform. reqs.             | <b>Req. 13.</b> The instrument shall have a resolution better than 1 unit.<br><b>Req. 14.</b> The instrument shall have a FOV greater than 2°.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Integrat. logistic support | <b>Req. 15.</b> The instrument GSE shall be compatible to instrument cleanliness requirements.<br><b>Req. 16.</b> The instrument contractor shall provide a transport container compatible with transport requirements to transport the instrument to the prime contractor facilities.<br><b>Req. 17.</b> The instrument EGSE shall provide no less than 400 W.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Physical reqs.             | <b>Req. 18.</b> The instrument shall fit within 1 m x 1 m x 1 m.<br><b>Req. 19.</b> The instrument shall have a mass lower than 900 kg.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Config. reqs.              | <b>Req. 20.</b> The instrument reference frame shall be according to RD02.<br><b>Req. 21.</b> The instrument MICD shall include the mass of the represented element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Design reqs.               | <b>Req. 22.</b> The instrument mass budget shall use the margin philosophy specified in RD01.<br><b>Req. 23.</b> The instrument shall use independent detectors for both spectral ranges.<br><b>Req. 24.</b> The instrument shall have a reliability higher than 0.90.<br><b>Req. 25.</b> The instrument shall be thermally decoupled from the platform.<br><b>Req. 26.</b> The instrument shall use redundant power supplies.<br><b>Req. 27.</b> The instrument shall provide venting provisions.<br><b>Req. 28.</b> The instrument shall be designed to operate in LEO.<br><b>Req. 29.</b> The instrument shall use a focal length of 3 units:<br>Note: in order to achieve resolution requirement Req. 13.<br><b>Req. 30.</b> The instrument shall use mirrors of XYZ material.<br><b>Req. 31.</b> The optical elements of the instrument shall be aligned with a precision of 10 units. |
| Environ. reqs.             | <b>Req. 32.</b> The instrument shall fulfill its performance requirements in the thermal environment specified in RD04.<br><b>Req. 33.</b> The instrument shall fulfill its performance requirements in vacuum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Verif. reqs.               | <b>Req. 34.</b> The instrument shall fulfill its performance after being subjected to 4 thermal cycles between -50°C and +100°C.<br><b>Req. 35.</b> The instrument performance shall be verified at 10 °C.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## 4. APPLICATION

### 4.1. Scope

A set of requirements for an optical space instrument is elicited and organized according to traditional categorization of requirements for space systems, in particular for this case following ECSS [2009]. These requirements are then transferred to the proposed NbC in this research, where requirements that are not relevant to the system are identified, as well as those that impose solution constraints without supporting the satisfaction of new needs. Finally, completeness of the allocation is evaluated. Table III lists the sample requirements to be used in this application example. It shall be noted that some of them can be considered bad requirements according to, for example, INCOSE [2012]; however, they reflect state-of-the-practice in space systems. The present research supports the identification of such requirements from a system definition perspective instead of traditional grammar rules check.

The given set of requirements corresponds to an extraction of what a set of requirements for an actual system would be, yet its content is sufficient to demonstrate the application of the proposed NbC. Acceptability and suitability of the sample requirements have been validated by deriving and contrasting them against requirements of actual operational and scientific optical space systems developed by different manufacturers for different customers and with a similar level of complexity, which is represented by an instrument mass of around 1 ton.

### 4.2. Use of Need-Based Categorization Model of Requirements (NbC)

The proposed model is not intended as an intermediate step to reorganize requirements, but as the basis to organize them during elicitation. Therefore, although the same requirements are used, this process should be understood as independent from that described in the previous section, i.e., Table III is not used as a basis to create Table IV.

Using the same elicitation techniques as in the previous section, Table IV shows requirements that are elicited and categorized according to the proposed NbC. Note that requirements Req. 36 to Req. 38 have been added to demonstrate the application of value levels. They are not mandatory, but a contractor may evaluate the potential benefit of fulfilling them. As discussed earlier, valuation is done on subsets of requirements that build on previous levels, thus providing always a complete definition of the system.

Table V shows why the missing requirements have been left out of the classification and consequently have not been elicited as system requirements.

Requirements identified as constraints could then be inserted in the *Resource* category in case a particular stakeholder would like to enforce a particular design decision into the system, at the expense of limiting the solution tradespace without fulfilling new needs.

### 4.3. Discussion

The set of requirements in the NbC defines a complete end product, consequently fulfilling the property of completeness

**Table IV. Sample Requirements Categorized According the Proposed NbC**

| Value level<br>to<br>stakeholders | Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                      |                                                                                                                                              |                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                   | Functions (Do)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Performance (Being)                                                                                                                                                                                                                  | Resources (Have)                                                                                                                             | Interaction (Interact)                                                                                                                                                                                                |
| Break-even                        | <b>Req. 5.</b><br>The instrument shall image the Earth surface in UV spectral range.<br><b>Req. 6.</b><br>The instrument shall image the Earth surface in VNIR spectral range.<br><b>Req. 8.</b><br>The instrument shall power itself.<br>Note: when receiving power according to Req. 12.<br><b>Req. 9.</b><br>The instrument shall transmit image data according to IF-Def-1.<br><b>Req. 11.</b><br>The instrument shall have a thermal coupling at its mechanical interface lower than 2 units.<br><b>Req. 12.</b><br>The instrument shall accept power according to IF-Def-2. | <b>Req. 13.</b><br>The instrument shall have a resolution better than 1 unit.<br><b>Req. 14.</b><br>The instrument shall have a FOV greater than 2°.<br><b>Req. 24.</b><br>The instrument shall have a reliability higher than 0.90. | <b>Req. 18.</b><br>The instrument shall fit within 1 m x 1m x 1 m.<br><b>Req. 19.</b><br>The instrument shall have a mass lower than 900 kg. | <b>Req. 32.</b><br>The instrument shall fulfill its performance requirements in the thermal environment specified in RD04.<br><b>Req. 33.</b><br>The instrument shall fulfill its performance requirements in vacuum. |
| Goal                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Req. 36</b><br>The instrument shall have a resolution better than 2 unit.                                                                                                                                                         | <b>Req. 37</b><br>The instrument shall have a mass lower than 750 kg.                                                                        |                                                                                                                                                                                                                       |
| Wish                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Req. 38</b><br>The instrument shall have a FOV greater than 5°.                                                                                                                                                                   |                                                                                                                                              |                                                                                                                                                                                                                       |

**Table V. Requirements Not Included as Systems Requirements in NbC**

| Justification for consideration as non-system requirement                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Req. 1.</b> Redundant requirement covered by Req. 5, Req. 6, and Req. 9. In addition, although the mission is to transmit images to ground the instrument only sends it to the platform, so the requirement is not verifiable.                                                                                                 |
| <b>Req. 2.</b> Redundant requirement covered by Req. 32 and Req. 33.                                                                                                                                                                                                                                                              |
| <b>Req. 3.</b> Operational requirement that does not depend solely on the instrument. Therefore impossible to verify. Functionality covered by Req. 8 and Req. 12.                                                                                                                                                                |
| <b>Req. 4.</b> Operational requirement that does not depend solely on the instrument and that imposes a solution decision. Therefore impossible to verify. Compatibility to depressurization defined by Req. 33. Design decision needed, which could lead to venting provisions or pressurization of the instrument, for example. |
| <b>Req. 7.</b> Functional constraint that imposes a design decision to use movable mirrors. It does not support the fulfillment of any new stakeholder need, as it is covered by Req. 5 and Req. 6.                                                                                                                               |
| <b>Req. 10.</b> Redundant requirement covered by Req. 18 and Req. 19.                                                                                                                                                                                                                                                             |
| <b>Req. 15.</b> Not a system requirement, but a GSE requirement. It should belong to a different set of requirements.                                                                                                                                                                                                             |
| <b>Req. 16.</b> Not a system requirement, but part of contractual obligations. It should be reflected in a different artifact.                                                                                                                                                                                                    |
| <b>Req. 17.</b> Not a system requirement, but a GSE requirement. It should belong to a different set of requirements.                                                                                                                                                                                                             |
| <b>Req. 20.</b> Not a system requirement, but to its documentation. It should be part of a different artifact, such as a Statement of Work for example.                                                                                                                                                                           |
| <b>Req. 21.</b> Not a system requirement, but to its documentation. It should be part of a different artifact, such as a Statement of Work for example.                                                                                                                                                                           |
| <b>Req. 22.</b> Not a system requirement, but to its documentation. It should be part of a different artifact, such as a Statement of Work or verification requirements.                                                                                                                                                          |
| <b>Req. 23.</b> Constraint that imposes a design decision to use independent detectors. It does not support the fulfillment of any new stakeholder need, as it is covered by Req. 5, Req. 6, Req. 13, Req. 14, and Req. 24.                                                                                                       |
| <b>Req. 25.</b> Redundant requirement covered by Req. 11.                                                                                                                                                                                                                                                                         |
| <b>Req. 26.</b> Constraint that imposes a design decision to use redundancy. It does not support the fulfillment of any new stakeholder need, as it is covered by Req. 24.                                                                                                                                                        |
| <b>Req. 27.</b> Constraint that imposes a design decision to use venting for depressurization. Compatibility to depressurization defined by Req. 33.                                                                                                                                                                              |
| <b>Req. 28.</b> Redundant requirement covered by Req. 32 and Req. 33.                                                                                                                                                                                                                                                             |
| <b>Req. 29.</b> Constraint that imposes a design decision on the optical path. It does not support the fulfillment of any new stakeholder need, as it is covered by Req. 13 and Req. 14.                                                                                                                                          |
| <b>Req. 30.</b> Constraint that imposes a design decision on the optical path. It does not support the fulfillment of any new stakeholder need, as it is covered by Req. 13 and Req. 14.                                                                                                                                          |
| <b>Req. 31.</b> Constraint that imposes a design decision on the optical path and integration activity. It does not support the fulfillment of any new stakeholder need, as it is covered by Req. 13 and Req. 14.                                                                                                                 |
| <b>Req. 34.</b> Verification requirements should be a different artifact and linked to each system requirement, in this case to Req. 32 and Req. 33.                                                                                                                                                                              |
| <b>Req. 35.</b> Verification requirements should be a different artifact and linked to each system requirement, in this case to Req. 13 and Req. 14.                                                                                                                                                                              |

of a partition in set theory: *Powered and commanded by an external system and launched by a particular rocket into Low Earth Orbit (LEO), the instrument takes images of the Earth in a specific spectral range with a specific performance and provides them to ground.* All of these elements form requirements that define what the system has to do, in which context, how well, and which resources it can use to achieve it. In fact,

every real system requirement has been seamlessly allocated to the categorization independently on what requirement type it represented in traditional categorizations. No system requirement has remained orphan.

An evaluation of the requirements in Table IV show that each requirement uniquely addresses one criterion, and there is no overlap between the different categories, consequently fulfilling the property of proper subsets in set theory. Yet the matrix definition reflects the dependencies between functional requirements and their associated performance, reflecting the actual dependency nature of some requirements at a given level of the architecture decomposition.

In addition, the needs that the system must satisfy have been defined with 13 requirements in the proposed NbC, while the use of a traditional categorization resulted in 35 requirements. Because the application of the categorization model has used an excerpt of requirements as an example, these figures do not provide a quantitative indication of the capability of the proposed NbC to reduce requirements. Yet they provide a qualitative indication of the general potential of using categorization models based on the concept of partition in set theory and, in particular, the potential capability of the proposed NbC in terms of requirement completeness and identification of constraints. These could eventually result in a maximization of the solution tradespace.

Scoping of system requirements to include only real requirements to the system and to exclude requirements to the project or any other ancillary information supports both the design and the procurement processes in various ways and can reduce uncertainties related to requirements management:

- Design efforts on each element can be streamlined by using a smaller set of requirements that in addition do not need applicability analysis.
- Procurement managers have a consolidated and domain-specific list of procurement requirements, mitigating the risk that some stay on the system requirements under responsibility and control of design engineers.
- Inconsistencies between flow-down and allocation of requirements, and system decomposition are minimized, as requirements to external systems are not included in the system requirements.
- Design efforts on each element can be streamlined by having a better separation between system and other deliverables.

Another interesting aspect of the proposed categorization model is its cognition property when used by requirements engineers to elicit, organize, and manage requirements, by stakeholders to understand how their needs are implemented in requirements, and by design engineers to transform requirements into boundary conditions of their solutions. Mapping requirements to how human needs are organized facilitate a natural perception of what needs to be accomplished. The organization of requirements in a matrix form instead of the traditional list approach improves visualization by understanding at a glance relations between different requirements. Without constraining the solution tradespace, such representation facilitates the understanding of requirement depend-

encies, thus improving the starting point of the exploration of the solution tradespace. This expanded capability in the use or requirements could therefore result in increased design efficiency and effectiveness.

Finally the organization of goals as packages that represent an *image* or a *view* of a system (in terms of need satisfaction) instead of presenting them as associations with individual requirements provides the capability to facilitate holistic decision making instead of local optimizations: Tradeoffs occur primarily between packages of needs instead of between individual performances. Efforts can therefore be planned from the perspective of value provided to stakeholders and not on the basis of optimizing dedicated performance attributes without evaluating their contribution to the overall system value.

## 5. CONCLUSIONS

The present paper begins with a review of existing categorizations of requirements that are used to support requirements elicitation, of existing representations and prioritizations of the value requirements have to stakeholders, and of psychological models of human needs. The paper presents a system-centric Need-based Categorization model of requirements that uses psychological models of human needs as a paradigm.

The proposed categorization model supports elicitation of requirements and facilitates the detection of requirements that are not relevant to the system and of constraints that limit the solution tradespace without supporting the satisfaction of new needs. In addition, the proposed categorization model organizes requirements in subsets that reflect value to stakeholders in a realistic manner and eases its usability by industry, consequently facilitating its adoption by practitioners. In conclusion, the proposed NbC model facilitates a complete definition of system boundaries with the minimum necessary amount of requirements that reflects value to stakeholder and consequently ultimately promotes system affordability.

The application of the proposed categorization model has been positively demonstrated for a set of sample requirements that reflect real categorization and requirements of actual optical space systems of medium to high complexity.

Finally the present research proposes future work on applying the proposed NbC model to full sets of requirements for large-scale systems in order to improve its validation and quantitatively assess the expected improvement in terms of reduction of requirements.

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